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G

H

NATIVE SOURCE. Power development; safety/PHY/PCB/bench qualification remain OPEN.

01_carrier_interface

10_input_protection

20_positive_rails

30_negative_reference

40_power_supervision

03_peripheral_boundaries

04_dac_0_3

05_dac_4_7

06_analog_outputs

41_power_status

File: 01_carrier_interface.kicad_sch

File: 10_input_protection.kicad_sch

File: 20_positive_rails.kicad_sch

File: 30_negative_reference.kicad_sch

File: 40_power_supervision.kicad_sch

File: 03_peripheral_boundaries.kicad_sch

File: 04_dac_0_3.kicad_sch

File: 05_dac_4_7.kicad_sch

File: 06_analog_outputs.kicad_sch

File: 41_power_status.kicad_sch

Sheet: /

File: servohl_io_revB.kicad_sch

Title: ServoHIL Rev.B – native editable single–SoC expansion

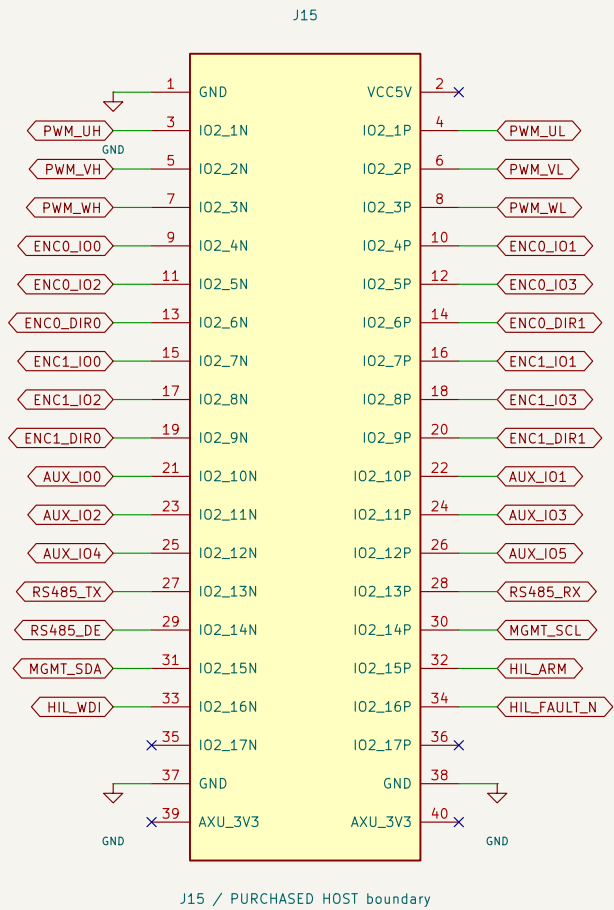
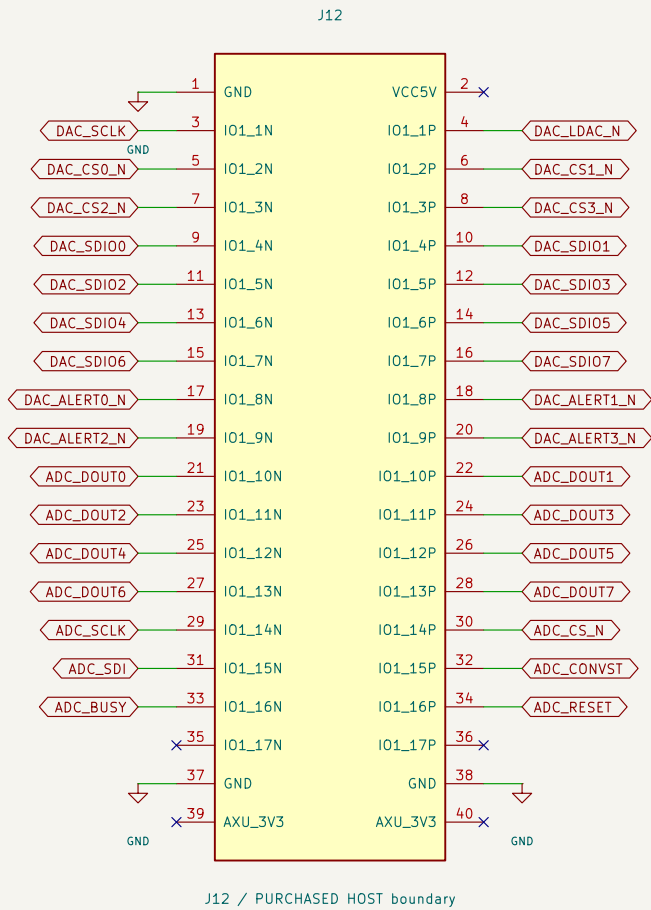
Size: A2

Date:

Rev: B–power–development

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Id: 1/11



Connector power contacts remain NC. 64 assigned GPIO; unused GPIO also NC. No secondary compute chip.

J12 and J15 are ORIGINAL AXU2CGB interfaces, not AXU2CGB-I or -E. No board-power backfeed through contacts.

Signal-pin backfeed, output enable and bank power sequencing STILL REQUIRE electrical review.

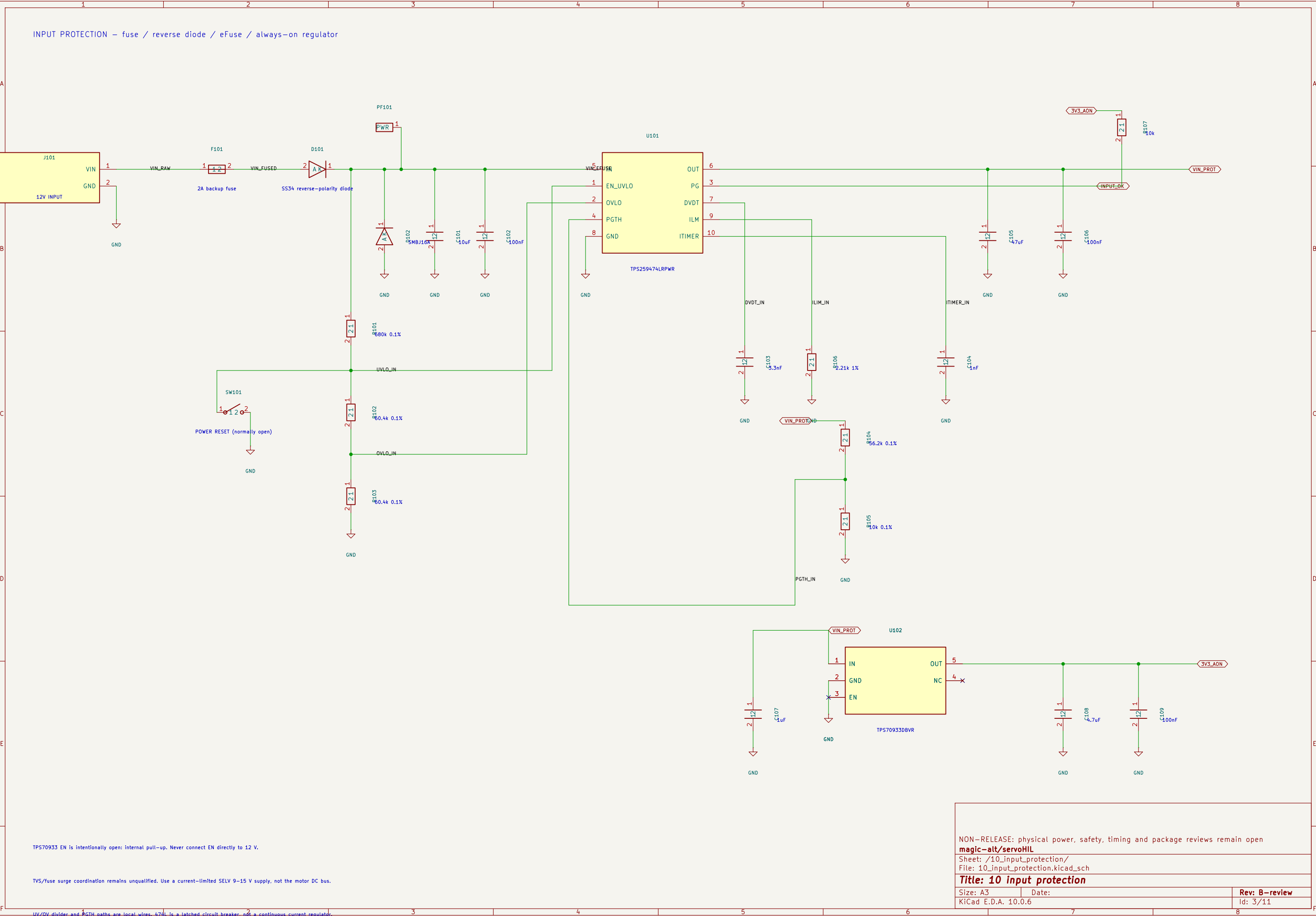
NON-RELEASE: physical power, safety, timing and package reviews remain open

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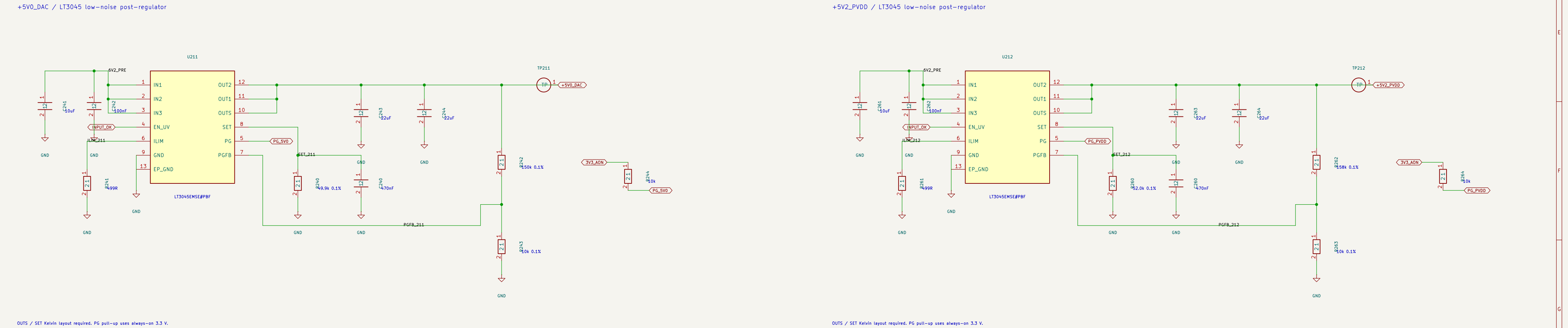
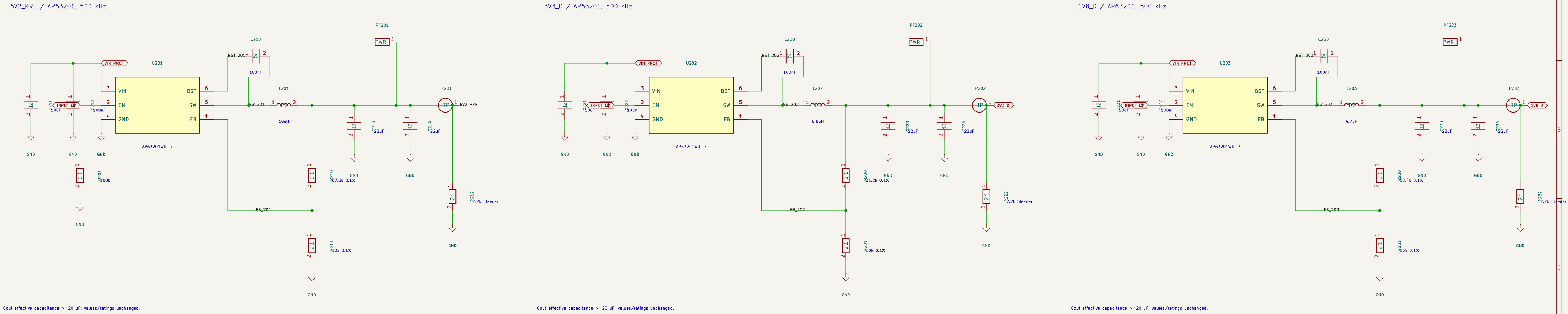
Sheet: /01_carrier_interface/
File: 01_carrier_interface.kicad_sch

Title: 01 carrier interface

Size: A3	Date:	Rev: B-review
KiCad E.D.A. 10.0.6		Id: 2/11



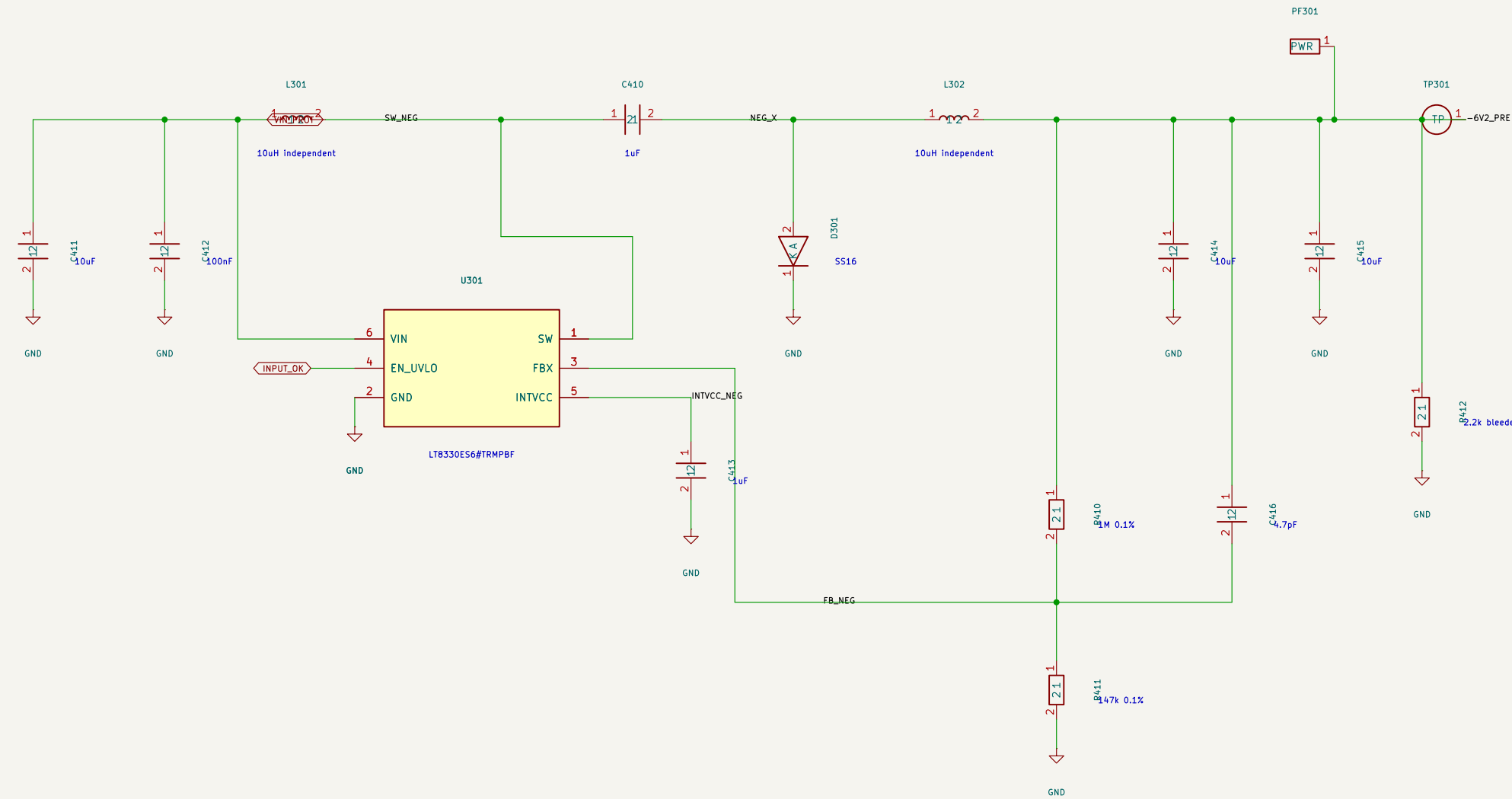
POSITIVE RAILS – direct power path, local feedback, cross-sheet interfaces only



All parts, pin numbers, values and footprints match the previous native circuit. These drawings do not establish bench qualification.

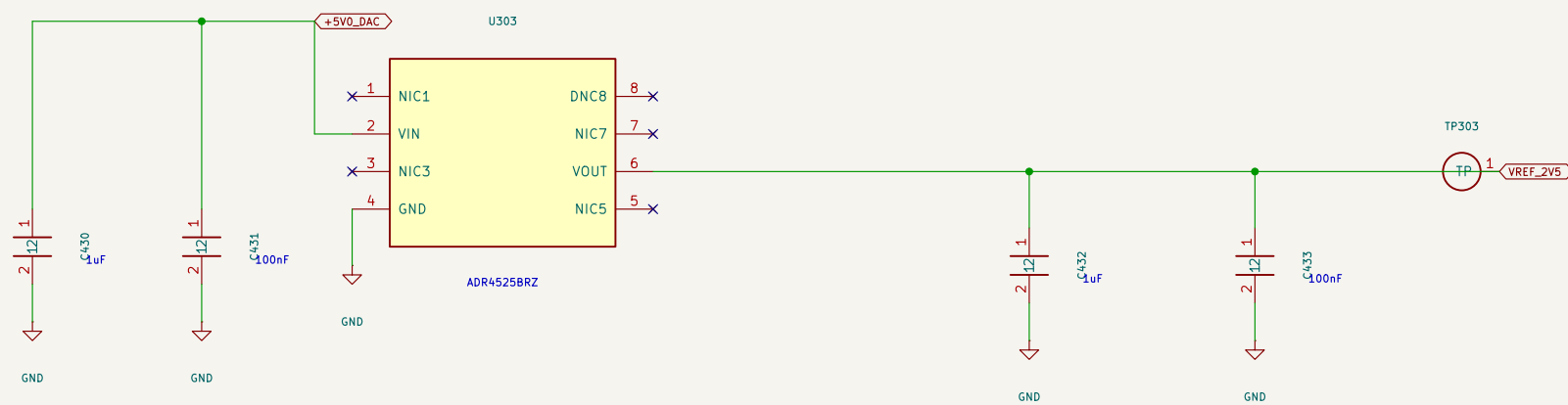
No second FPGA. Output current, capacitor DC bias, startup, ripple, thermal and safety behavior remain hardware review gates.

NEGATIVE RAIL / REFERENCE – wired Cuk converter, LT3094 and ADR4525



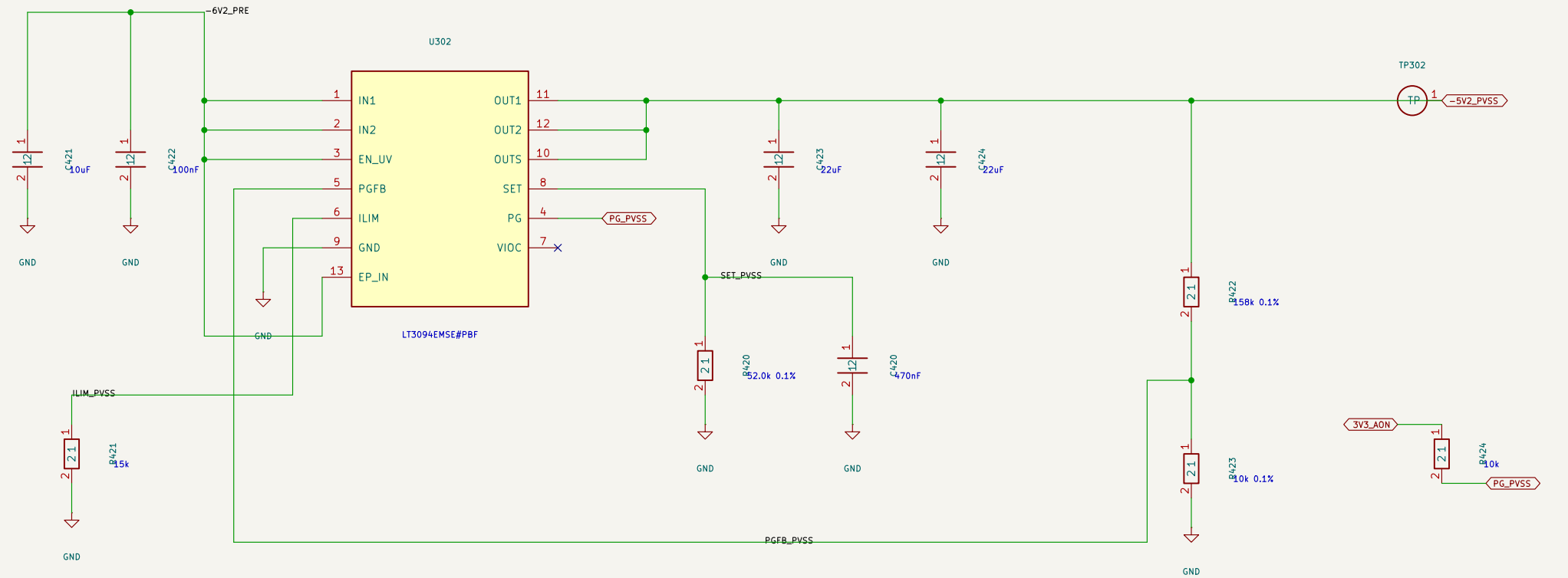
Cuk current path: VIN -> L301 -> C410 -> L302 -> negative output. D301 cathode is grounded.

Independent inductors; transfer capacitor sees $V_{IN} + |V_{OUT}|$. Existing component values are preserved.



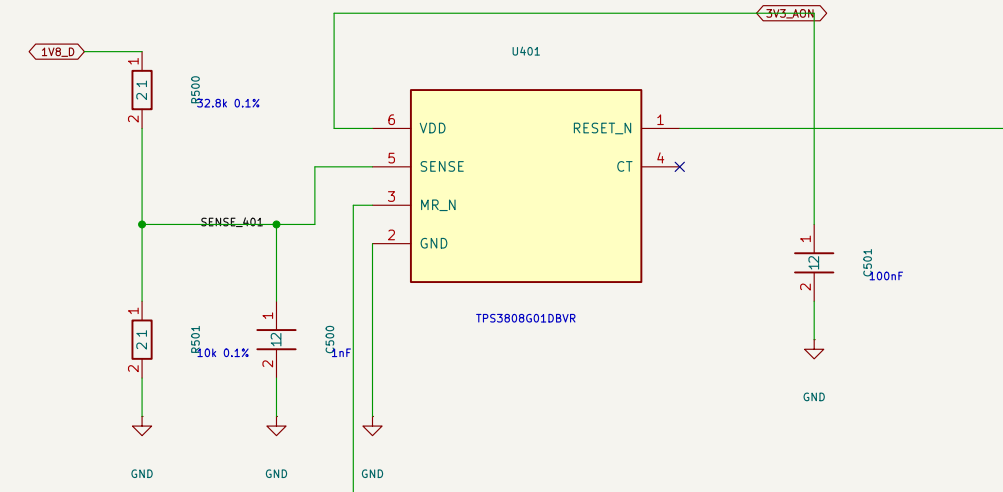
LT3094 EP13 / EN3 connect to negative IN. OUTS/SET require Kelvin layout. No startup/ripple/thermal PASS is implied.

ADR4525: unused NIC/DNC pads remain NC. DAC external—reference configuration and actual fanout/load require validation.

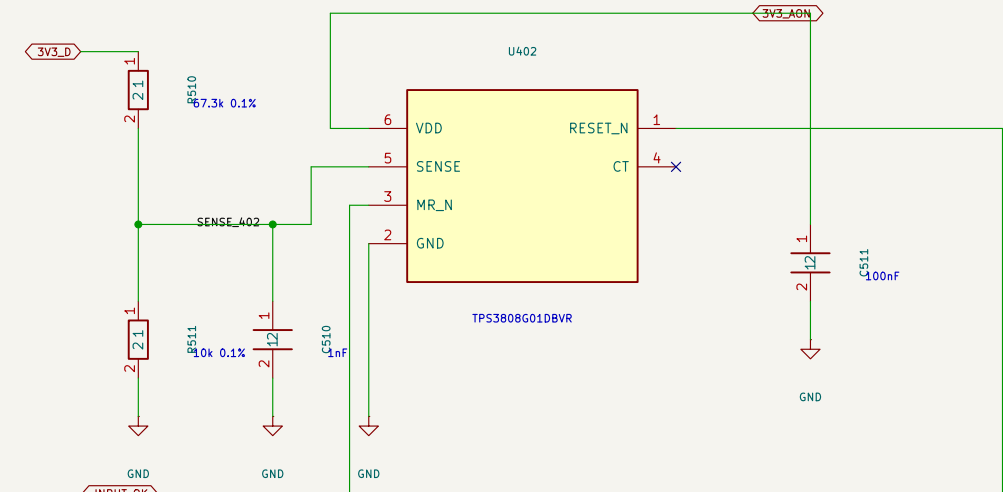


POWER QUALIFICATION – rail monitors, negative sense, deliberate re–arm and reset driver

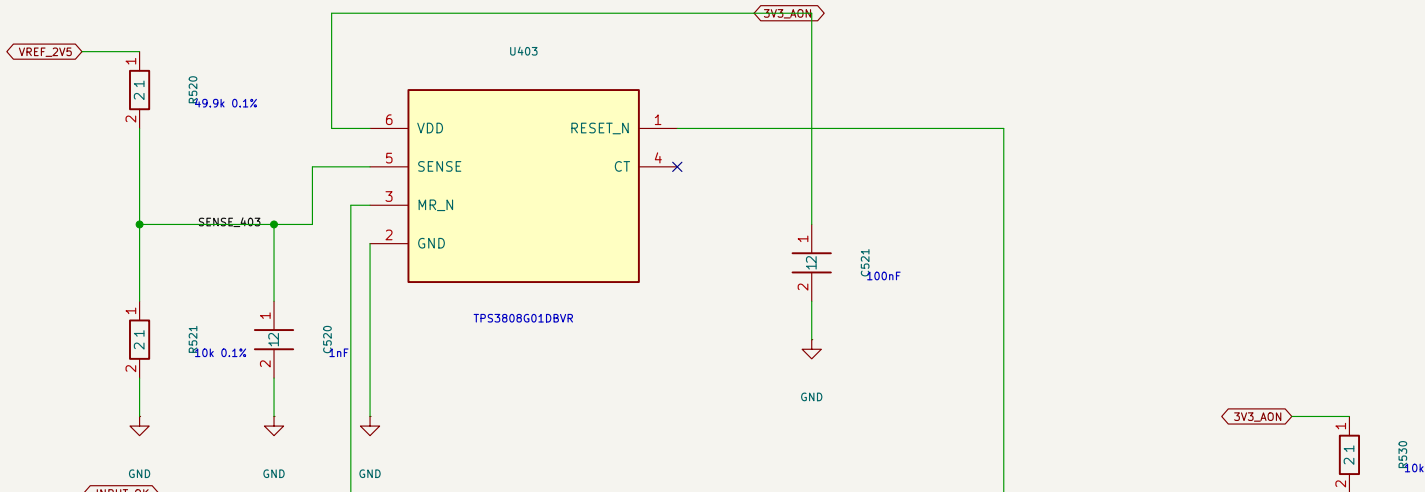
1V8_D rail supervision



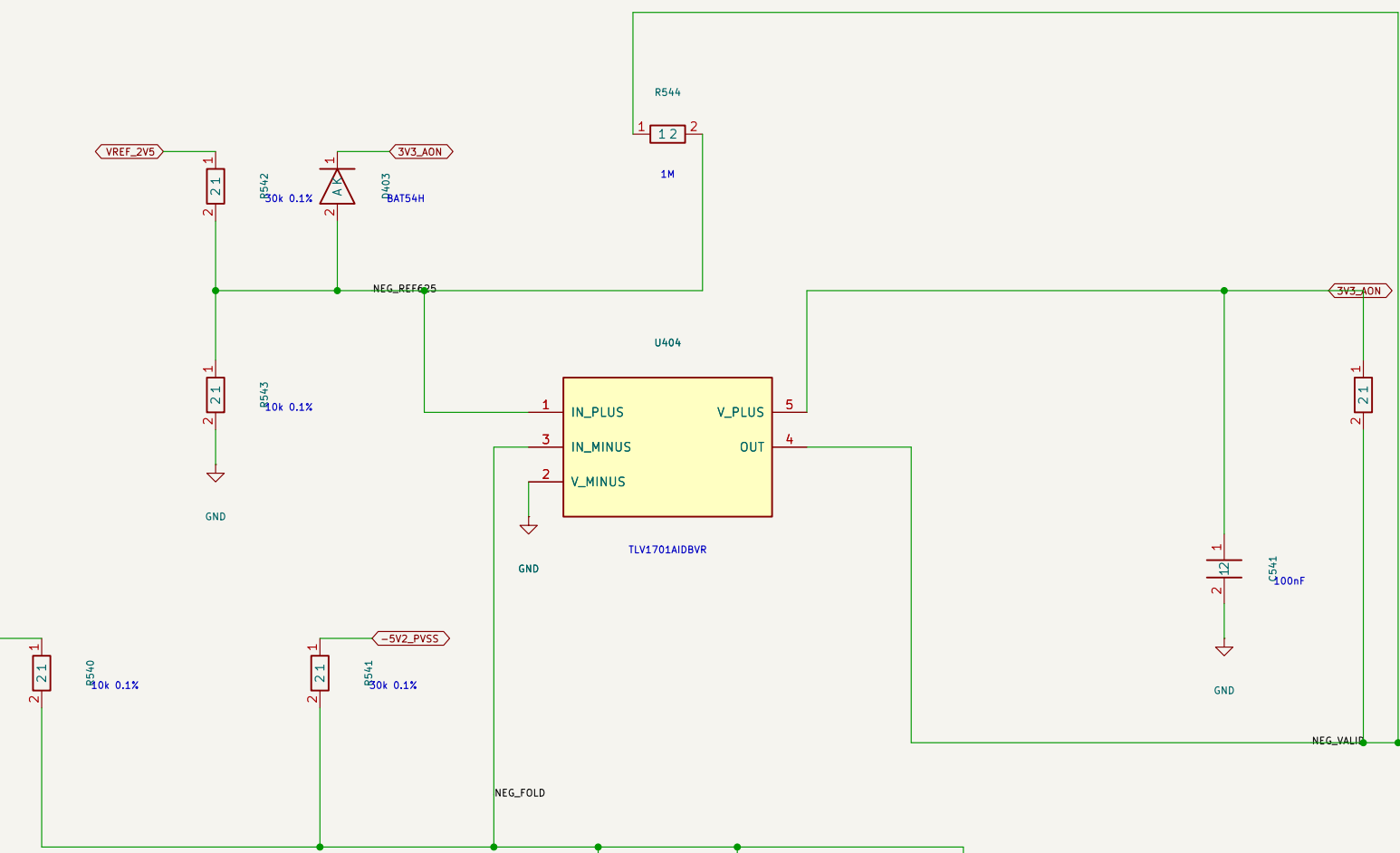
3V3_D rail supervision



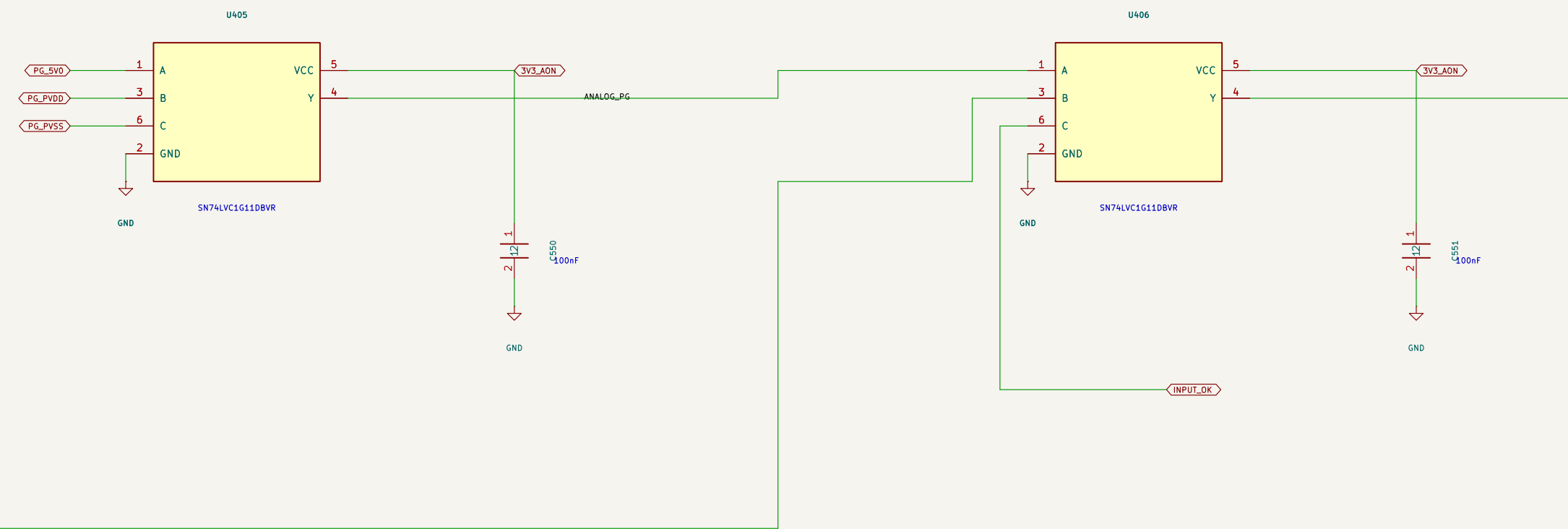
VREF_2V5 rail supervision



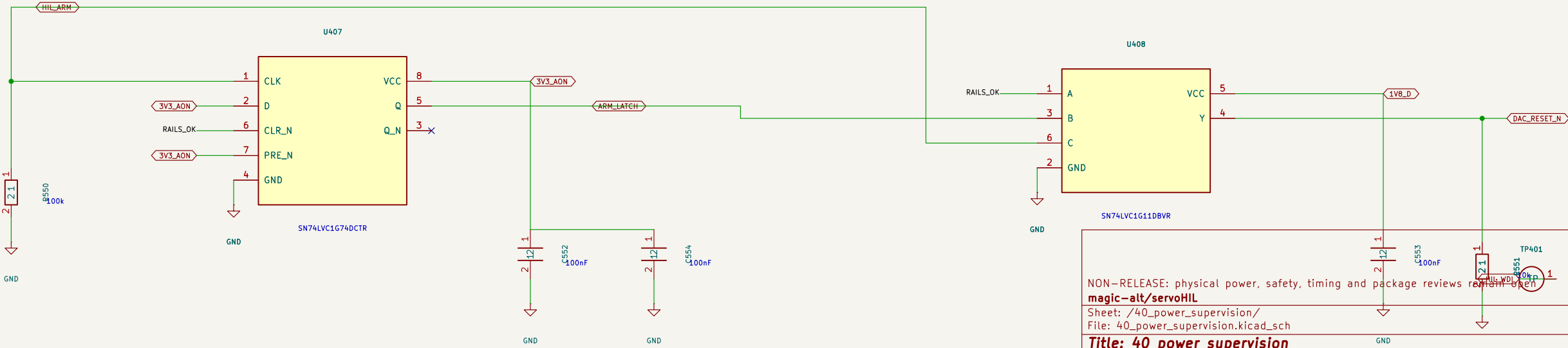
PVSS independent sense / hysteresis



Supply qualification



Re–arm latch / immediate disarm

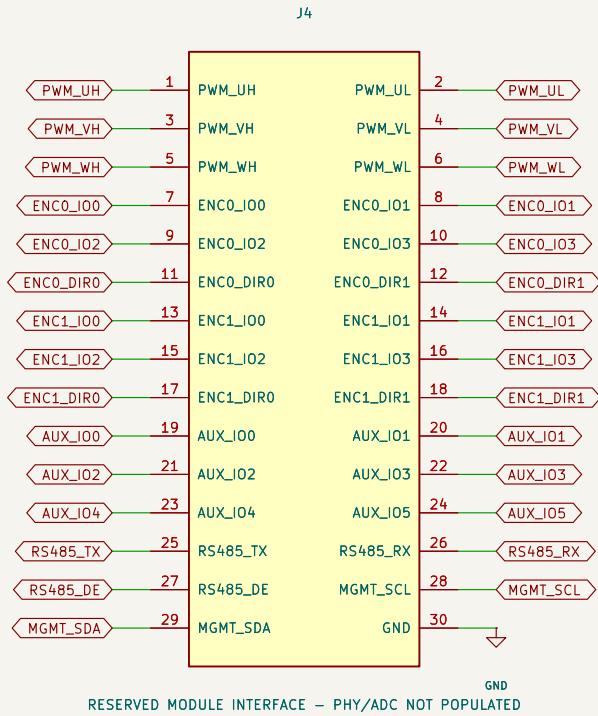
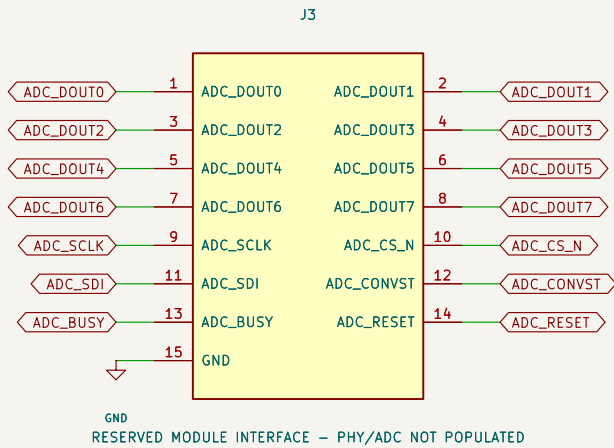


TPA01_HIL_WDI is a reserved Input. NOT a watchdog. Complete heartbeat, AO disconnect and DUT gate inhibit remain OPEN.

Power fault clears the latch: a fresh ARM edge is required. HIL_ARM LOW immediately holds DAC reset asserted. Hardware validation remains pending.

NON–RELEASE: physical power, safety, timing and package reviews remain pending.
magic–alt/servoHIL
Sheet: /40_power_supervision/
File: 40_power_supervision.kicad_sch
Title: 40 power supervision
Size: A2 Date: Rev: B–review
KiCad E.D.A. 10.0.6 Id: 6/11

03 – ADC and digital PHY integration interfaces (reserved)



Eight AI channels are an interface reservation. No 1 MSPS verification is implied; 1.8V serial timing must be closed.

Each encoder uses one selected profile; ABZ/SSI/BISS/SPI are NOT simultaneously independent ports.

NON–RELEASE: physical power, safety, timing and package reviews remain open

magic–alt/servoHIL

Sheet: /03_peripheral_boundaries/

File: 03_peripheral_boundaries.kicad_sch

Title: 03 peripheral boundaries

Size: A3

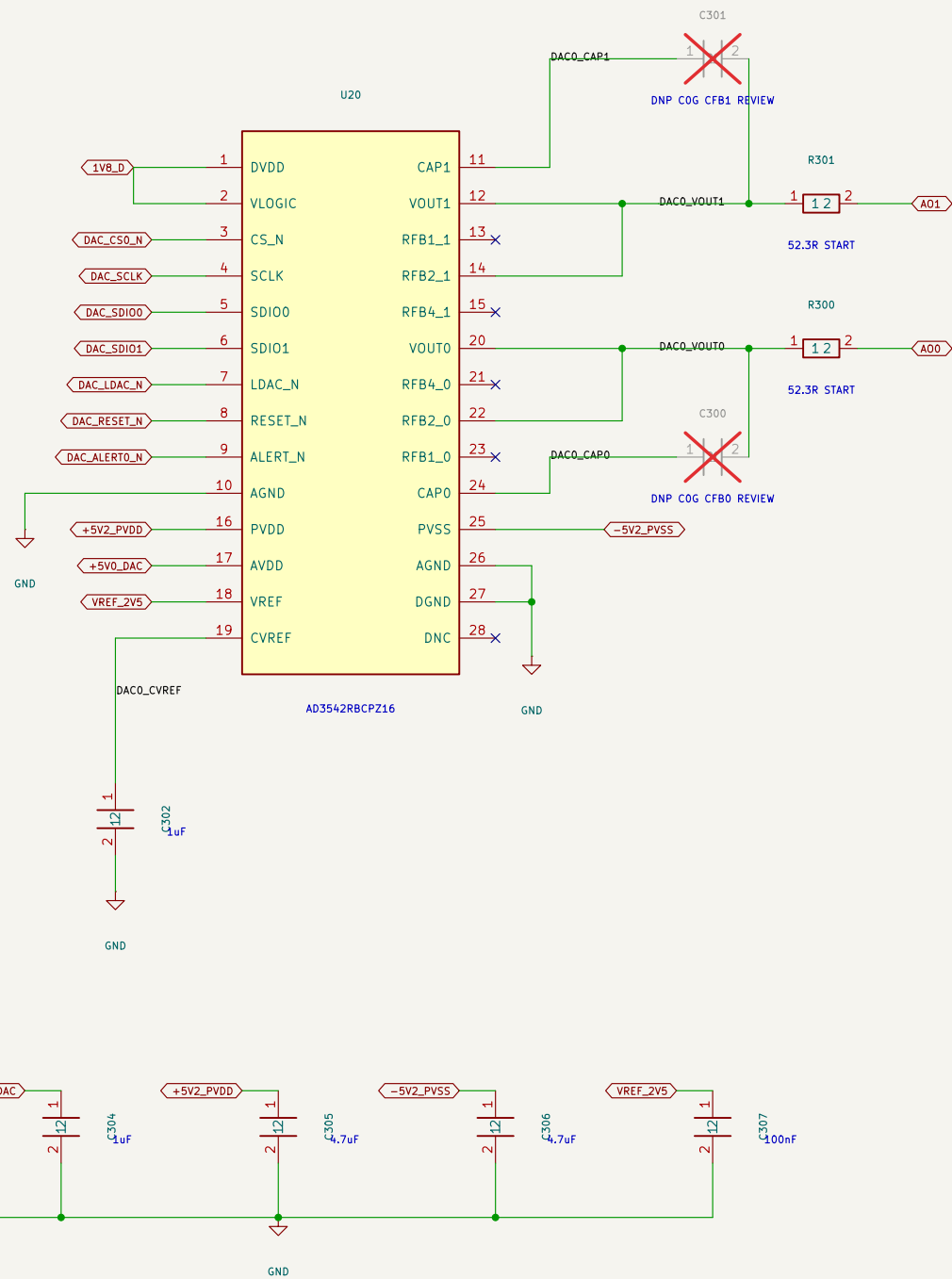
Date:

Rev: B–review

KiCad E.D.A. 10.0.6

Id: 7/11

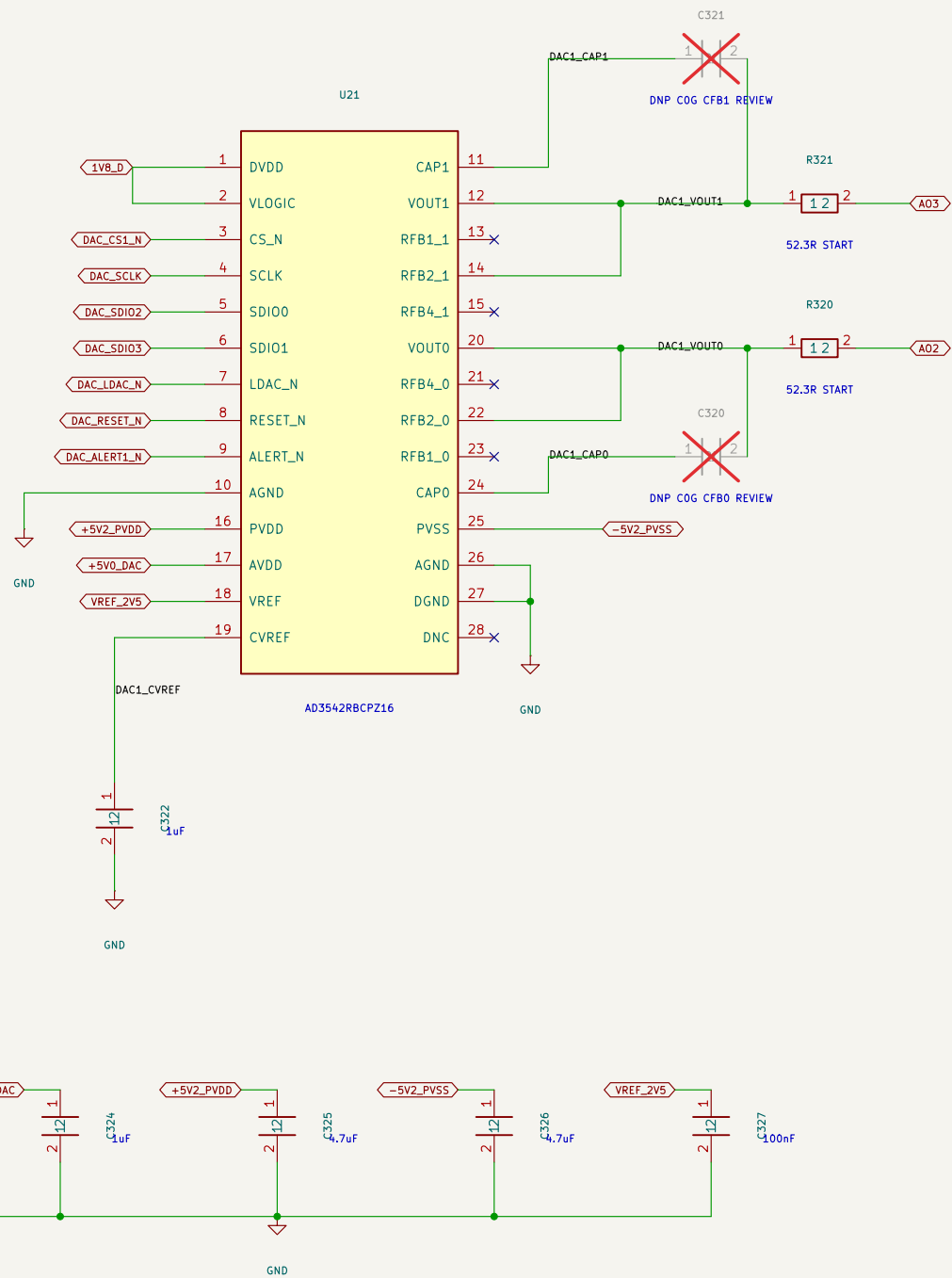
DAC – direct RFB2 feedback / CFB / output resistors; independent channels preserved



U20: A00 / A01. CFB remains DNP pending analog stability review.

Only inter-sheet signals use global labels. Feedback and output networks are drawn as visible closed local connections.

No electrical component/value change. CFB, output loading, package land pattern and hardware-safe AO behavior remain OPEN.



U21: A02 / A03. CFB remains DNP pending analog stability review.

NON-RELEASE: physical power, safety, timing and package reviews remain open

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Sheet: /04_dac_0_3/

File: 04_dac_0_3.kicad_sch

Title: 04 dac 0 3

Size: A3

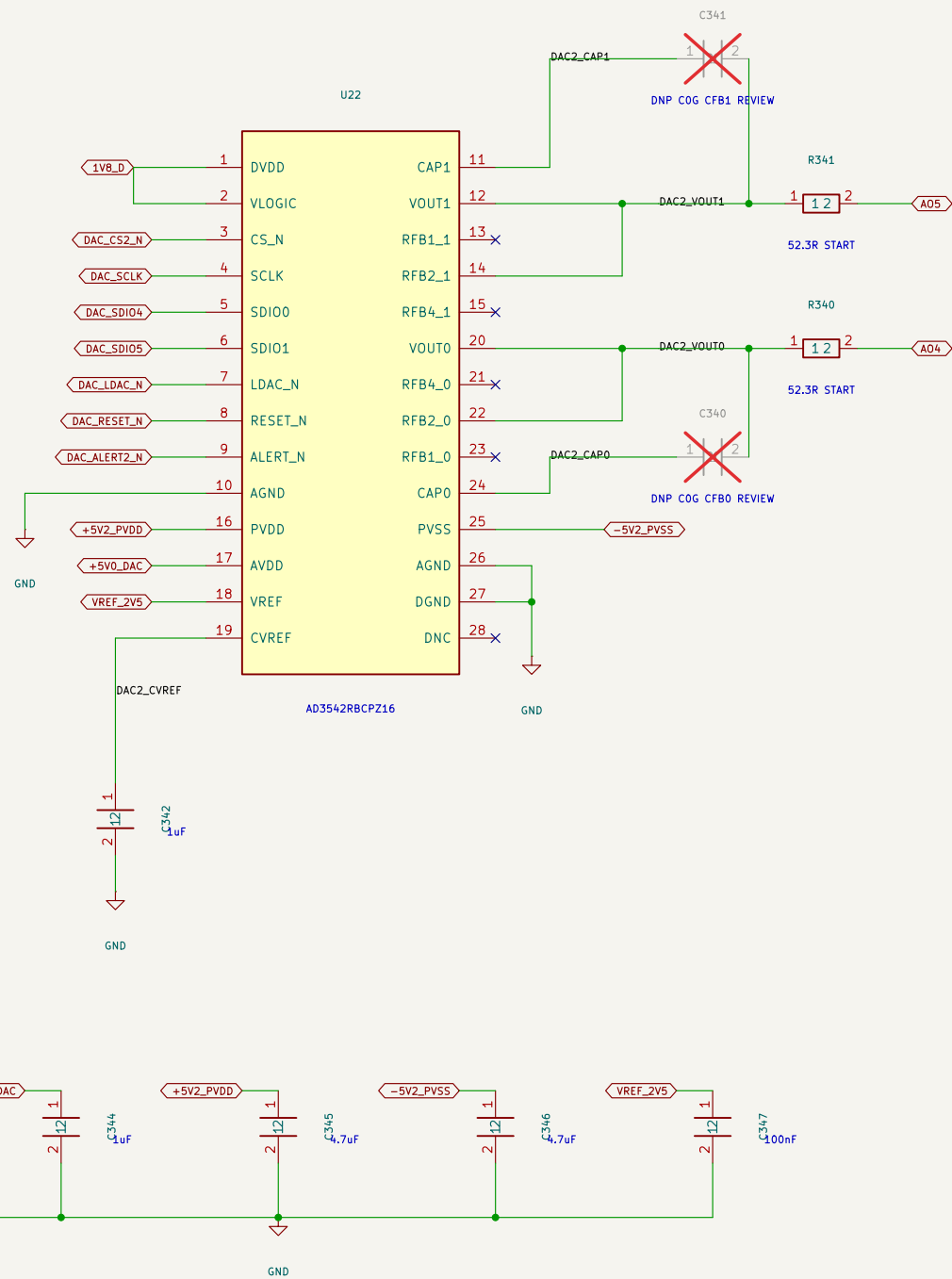
Date:

Rev: B-review

KiCad E.D.A. 10.0.6

Id: 8/11

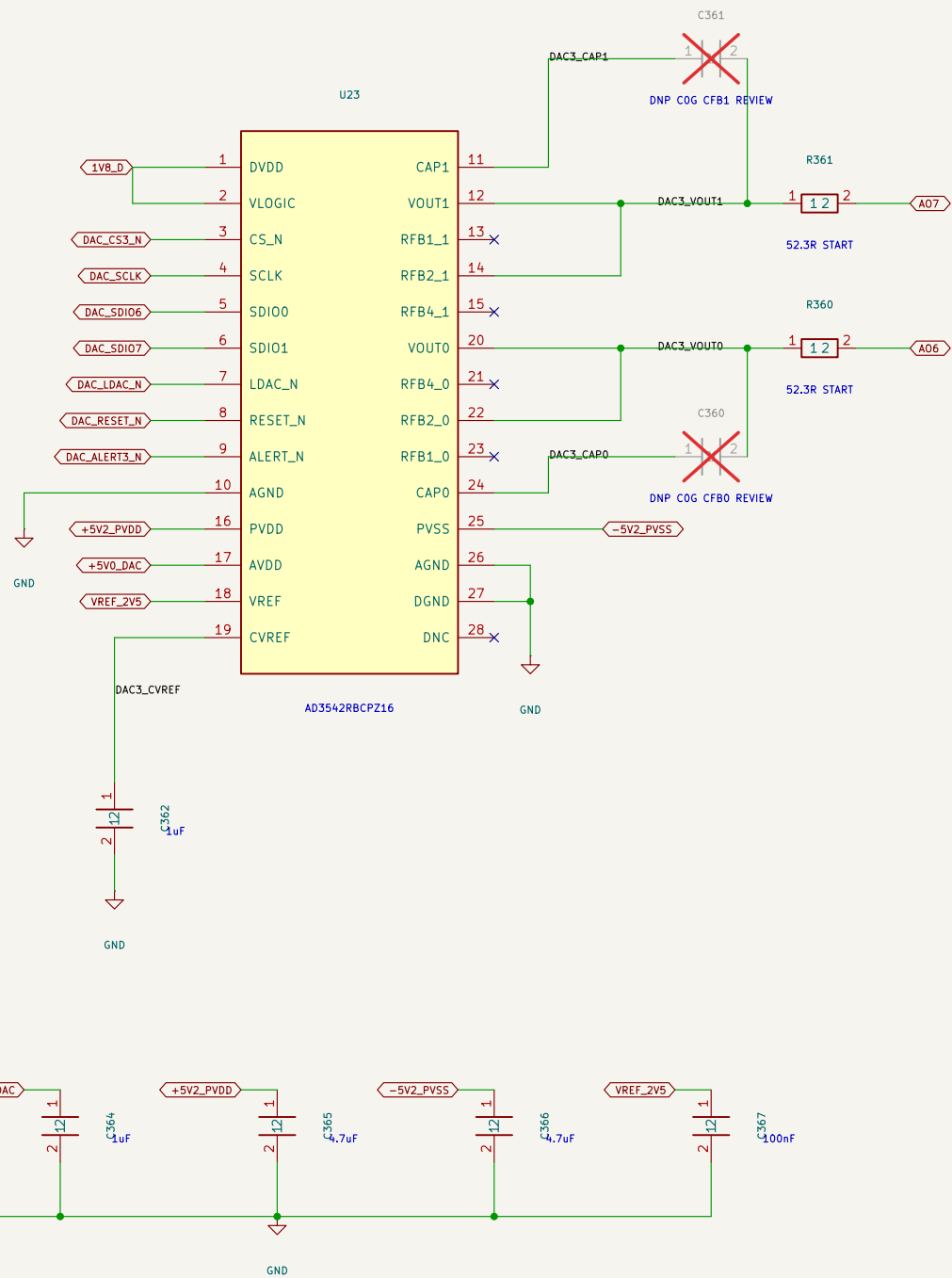
DAC – direct RFB2 feedback / CFB / output resistors; independent channels preserved



U22: A04 / A05. CFB remains DNP pending analog stability review.

Only inter-sheet signals use global labels. Feedback and output networks are drawn as visible closed local connections.

No electrical component/value change. CFB, output loading, package land pattern and hardware-safe AO behavior remain OPEN.



U23: A06 / A07. CFB remains DNP pending analog stability review.

NON-RELEASE: physical power, safety, timing and package reviews remain open

magic-att/servoHIL

Sheet: /05_dac_4_7/

File: 05_dac_4_7.kicad_sch

Title: 05 dac 4 7

Size: A3

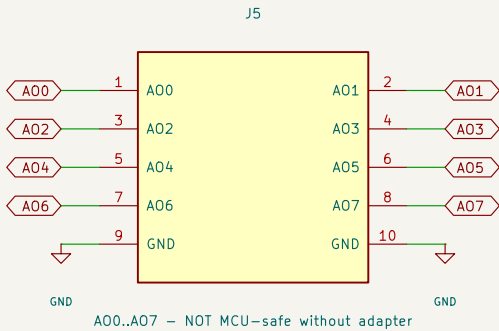
Date:

Rev: B-review

KiCad E.D.A. 10.0.6

Id: 9/11

06 – Eight analog outputs to reviewed DUT adapter



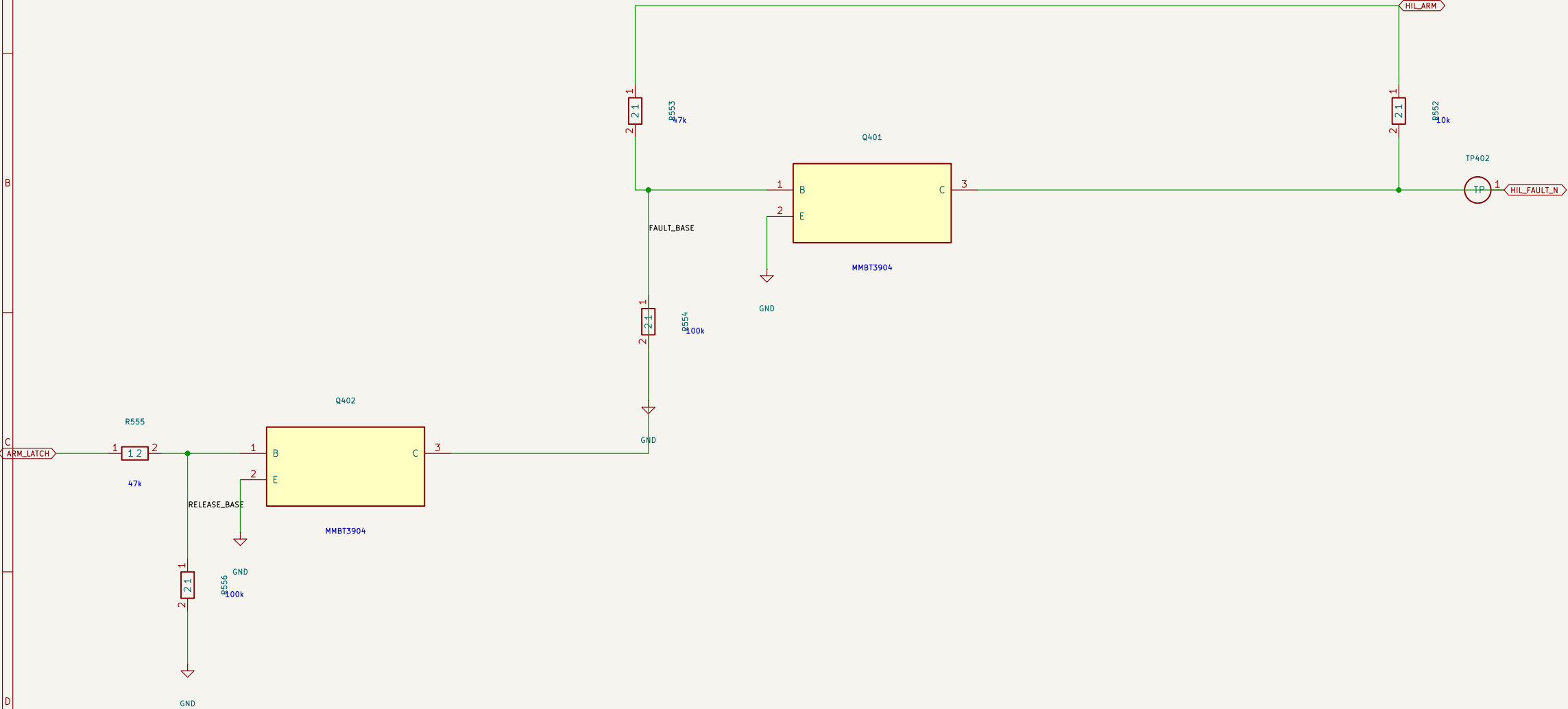
Default metadata: Ia, Ib, Ic, Vbus, torque, temperature, AUX0, AUX1. Electrical channels are equivalent.

DUT adapter supplies scaling/clamping and hardware inhibition. No direct +/-5V connection to MCU ADC.

NON–RELEASE: physical power, safety, timing and package reviews remain open
~~magic–alt/servoHIL~~
Sheet: /06_analog_outputs/
File: 06_analog_outputs.kicad_sch

Title: 06 analog outputs		
Size: A3	Date:	Rev: B–review
KiCad E.D.A. 10.0.6	Id: 10/11	

POWER STATUS – host–referenced default fault and qualified release



HOST HIL_ARM supplies both the status pull–up and default–fault bias. Expansion supply never directly pulls the status high.

ARM_LATCH drives Q402: only qualified, armed power releases Q401. HIL_FAULT_N LOW also means disarmed.

This is power status, not a full watchdog or certified safety function. Partial–power/transient behavior still requires bench tests.

NON–RELEASE: physical power, safety, timing and package reviews remain open

magic–alt/servoHIL

Sheet: /41_power_status/

File: 41_power_status.kicad_sch

Title: 41 power status

Size: A3

Date:

Rev: B–review

KiCad E.D.A. 10.0.6

Id: 11/11